Hong Ong

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Professional Summary

Software Engineering graduate (First Class Honours, UTS) with nearly a year of production experience. Sole software engineer on OnlyFacts' Ruby on Rails/PostgreSQL climate data platform, and developer on The a2 Milk Company's multi-market Astro/TypeScript websites. Full-stack across data pipelines, APIs, CMS-driven components, localisation and automated testing, with hands-on experience maintaining CI/CD pipelines.

Work Experience

- **The a2 Milk Company** — Sydney, NSW
  Software Developer (Contract) — *June 2026 – Present*
  - Developed features for multi-market Astro 5 SSR sites (AU, NZ, SG, HK, VN, CA, US, China) with TypeScript, Kontent.ai (headless CMS), Alpine.js, Algolia and Tailwind CSS on AWS
  - Shipped 125+ commits across 3 production repositories, delivering 12+ reusable CMS-driven components with documentation and Vitest unit and smoke tests
  - Built a modular article system of 7 CMS-driven components for a health-professional research hub, launching 6 articles to production
  - Localised sites for English, Vietnamese and Chinese audiences with a tested UI-label translation utility, and integrated Osano cookie consent and GTM
- **OnlyFacts** — Sydney, NSW
  Full-Stack Engineer (Contract) — *January 2026 – Present*
  - Managed all engineering on the Ruby on Rails/PostgreSQL climate data platform as sole software engineer since mid-2026, working with a 4–6 person data team
  - Replaced Highcharts maps with a custom D3.js engine (council-area and world maps, hexbin view, zoom, legend) and refactored 1,300+ lines of legacy code
  - Developed data pipelines: web scrapers (vehicle sales, carbon credits) and recurring import jobs for emissions and energy datasets
  - Shipped a Cloudflare Worker contact API (validation, Google Sheets logging, email alerts, CORS allowlist) and GA4 event tracking for the live React/Vite OnlyFacts Studio site

  Software Engineering Intern — *November 2025 – January 2026*
  - Built and presented 3 feature prototypes to stakeholders, which informed later architecture decisions
  - Identified technical debt and maintainability issues in the legacy codebase and presented improvement recommendations adopted in later development
- **Optik Consultancy (Philips Dynalite)** — Sydney, NSW
  Software Engineering Consultant — *May 2024 – February 2025*
  - Led a 4–5 person team of software and mechanical engineers to deliver client automation solutions on time and within budget
  - Programmed OMRON TM5 cobot using TM FLOW to automate relay/terminal placement, cutting production time by 40% and error rates by 60%

Education

- **University of Technology Sydney** — Sydney, NSW
  Bachelor of Engineering (Honours), Software Engineering — *February 2022 – December 2025*
  - First Class Honours | WAM: 87.10 (High Distinction Average) | GPA: 6.59/7.0 | Dean's List (3 consecutive years) | Undergraduate Academic Excellence International Scholarship
  - Honours Capstone (HD, showcase nominee): built a C++/Python AST-based plagiarism detector that flagged 8 confirmed misconduct cases over 406 submissions, 73% faster with 73% less storage than text matching
  - Relevant Coursework: Database Fundamentals (100), Cloud Computing & SaaS (97), Software Innovation Studio (96), Software Analysis Studio (94), Software Architecture (93), Systems Testing and Quality Management (89)

Projects

- **VR Career Exploration: Lifeguard Simulation** August 2025 – December 2025
  Lead Programmer & VR Developer
  - Built immersive VR training simulation in Unity/C# using XR Interaction Toolkit; implemented a VR controller-based selection system with haptic, audio, and visual feedback cues, achieving 95% selection accuracy among first-time users during a live showcase
  - Developed a dynamic NPC system with randomised spawning; optimised performance for VR headsets using object pooling and reduced draw calls during NPC interactions, cutting frame drops by roughly 50% measured via Unity profiler
  - Led a 7-person Agile team, conducting code reviews and technical QA across 3D modelling, animation, and UI work
- **Thrive X Mobile App Prototype** February 2025 – April 2025
  Full-Stack Developer
  - Built a cross-platform mobile app in Flutter (Dart) integrating Google Gemini API to power AI-driven skill assessments and personalised feedback, spanning front-end, REST APIs, Firebase, and AI workflow orchestration
  - Implemented real-time data sync via Firebase (30% faster load times), RESTful API integration, and user authentication/profile management

Leadership Activities

- **UTS Malaysian and Singaporean Students' Association (MASSA)** Sydney, NSW
  Treasurer *November 2022 – June 2024*
  - Managed $40,000+ budget across 20+ events, consistently achieving break-even or profit targets through strategic financial planning and resource allocation

Technical Skills

- **Programming Languages**: JavaScript, TypeScript, Ruby, Python, SQL, Java, C, C#, C++
- **Frameworks**: Ruby on Rails, ReactJS, Node.js, Astro, Alpine.js, Flutter, D3.js, Tailwind CSS
- **CMS & Search**: Kontent.ai (headless CMS), Algolia
- **APIs & Architecture**: RESTful API design & integration, MVC, object-oriented programming (OOP), background job processing (Heroku Scheduler)
- **Databases**: PostgreSQL, MySQL, Firebase Realtime Database
- **Cloud**: AWS (CloudFront, Elastic Beanstalk, CodeBuild, S3), Google Cloud Platform (GCP)
- **DevOps**: Git, GitHub, Docker, CI/CD pipelines (GitHub Actions, AWS CodeBuild), Agile/Scrum, VS Code, JetBrains Suite
- **Testing**: Unit & integration testing, test-driven development (TDD); pytest, Google Test (C++) from coursework and personal projects
- **AI & Developer Tooling**: Google Gemini, Claude (Claude Code), AI workflow integration, GitHub Actions automation

Languages

- **Spoken Languages**: English, French (Fluent); Vietnamese, Mandarin, Cantonese (Basic)